



Chandigarh Engineering College Jhanjeri
Mohali-140307
Department of Artificial Intelligence (AI) and Data Sciences

IoT-Based Patient Health Monitoring System

Engineering Clinics – Multi Disciplinary Project
ECS-201

BACHELOR OF TECHNOLOGY
(Artificial Intelligence and Data Science.)



SUBMITTED BY:

Aditya Arora - 2420640
Abhishek Kumar - 2420636
Abhishek Giri - 2420637
Nov 2025

Under the Guidance of

Mrs. Manmeet Kaur
Assistant of department
CSE - APEX

**Department of Artificial Intelligence and Data Science
Chandigarh Engineering College Jhanjeri Mohali - 140307**



Chandigarh Engineering College Jhanjeri
Mohali-140307
Department of Artificial Intelligence (AI) and Data Sciences

DECLARATION

We, Aditya Arora, Abhishek Kumar, Abhishek Giri hereby declare that the report of the project entitled “Patient Health Monitoring System” has not presented as a part of any other academic work to get my degree or certificate except Chandigarh Engineering College Jhanjeri, Mohali, affiliated to I.K. Gujral Punjab Technical University, Jalandhar, for the fulfillment of the requirements for the degree of B. Tech in Artificial Intelligence and Data Science

Aditya Arora – 2420640

Mrs. Manmeet Kaur

Abhishek Kumar – 2420636

Assistant of department

Abhishek Giri – 2420637

CSE - APEX

Signature of the Head of Department
(with Stamp)



Chandigarh Engineering College Jhanjeri

Mohali-140307

Department of Artificial Intelligence (AI) and Data Sciences

ACKNOWLEDGEMENT

It gives us great pleasure to deliver this report, we worked on for our B. Tech in Artificial Intelligence and Data Science 3rd semester, which was titled “Patient Health Monitoring System”. We are grateful to our university for presenting us with such a wonderful and challenging opportunity. We also want to convey our sincere gratitude to all coordinators for their unfailing support and encouragement.

We are extremely thankful to the HOD and Project Coordinator of CSE – APEX Department at Chandigarh Engineering College Jhanjeri, Mohali (Punjab) for valuable suggestions and heartiest co-operation.

We are also grateful to the management of the institute for giving us the chance to acquire the information. We are also appreciative of all our faculty members, who have instructed us throughout our degree.

Aditya Arora – 2420640

Abhishek Kumar – 2420636

Abhishek Giri - 2420637



Table of Contents

S.No.	Contents	Page No
1.	Introduction	5
2.	System Requirements	6
3.	Software Requirement Analysis	7
4.	Software Design	8
5.	Implementation	9
6.	Results and Discussions	10-11
7.	Conclusion & Future Scope	12
8.	References	13



Chapter 1

Introduction

1.1 Project Introduction

The **IoT-based Patient Health Monitoring System** is designed to continuously monitor patients' vital health parameters such as **heart rate and blood oxygen level (SpO2)**. This system is particularly useful in hospitals, clinics, and home-based healthcare to track patient health in real-time. The system uses the **MAX30100 or MAX30102 sensor** for measuring heart rate and SpO2, displays readings on an **I2C LCD**, and transmits data for further analysis using **Arduino Uno R3 [6][8]**.

1.2 Objectives of the Project

- To monitor **real-time heart rate and SpO2 levels** of a patient.
- To display patient health data on an **LCD** for easy visualization.
- To familiarize with **IoT concepts**, sensors, and microcontroller interfacing.
- To provide a **cost-effective, portable, and efficient** solution for health monitoring.

1.3 Tools Learned

- **Arduino Uno R3:** Programming and hardware interfacing.
- **MAX30100/MAX30102 sensor:** Pulse oximeter and heart rate sensing.
- **I2C LCD Display:** Displaying real-time health parameters.
- **Arduino IDE:** Writing and uploading code to the microcontroller.
- **Basic IoT Concepts:** Sensor data collection, real-time monitoring.



Chapter 2

System Requirements

2.1 Hardware Requirements[3][5]



- Arduino Uno R3



- MAX30100 / MAX30102 Sensor



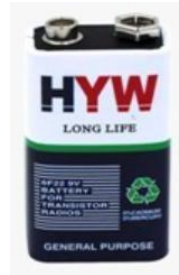
- I2C LCD Display (16x2 or 20x4)



- Breadboard and connecting wires



- USB Cable for Arduino programming



- Power supply (5V)

2.2 Software Requirements[3][5]

- Arduino IDE (latest version)
- MAX30100/MAX30102 Arduino library
- I2C LCD library (LiquidCrystal_I2C)
- PC for programming and monitoring
- C++ Programming Language



Chapter 3

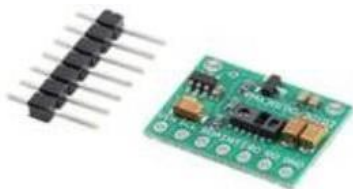
Software Requirement Analysis

3.1 Problem Definition

The proposed **IoT-Based Patient Health Monitoring System** addresses this need by using **sensors and microcontrollers** to automatically monitor and display vital signs such as **heart rate** and **SpO₂** without constant human supervision. The system leverages the **MAX30100/MAX30102** pulse oximeter sensor [1][7][9] for data collection, controlled through an **Arduino Uno** microcontroller [2]. This approach aligns with prior IoT-based remote health solutions proposed by Arefin and Azad [6] and Vasam et al. [8], who demonstrated that IoT integration enhances accessibility and efficiency in patient care.

3.2 Module Division and Functionalities

<u>Module Name</u>	<u>Description</u>
Sensor Module	Collects heart rate and SpO ₂ data using the MAX30100/MAX30102 sensor [1][7][9]. The sensor operates based on photoplethysmography (PPG) to detect oxygen saturation and pulse rate.
Processing Unit	Arduino Uno processes sensor data and formats it
Display Module	I2C LCD module displays BPM and SpO ₂ values in real time [4].
Optional Future Module	Wireless module (e.g., ESP8266) to transmit data to cloud/mobile app[5][6][8].



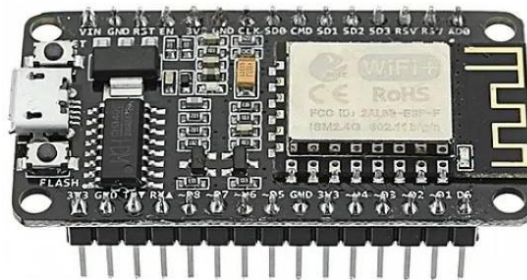
• Sensor Module



• Processing Unit



• Display Module

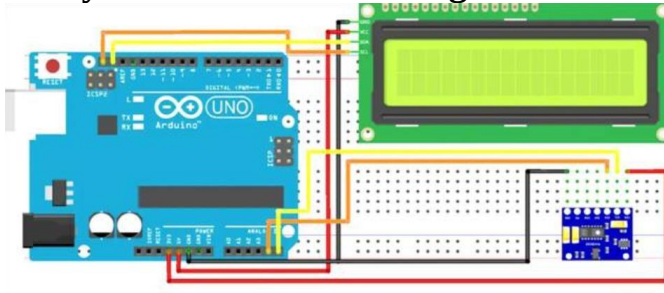


• Optional Future Module

Chapter 4

Software Design

4.1 System Architecture Diagram



4.2 Wiring Implementation

MAX30102 Sensor to Arduino Uno:

VCC → 3.3V or 5V on Arduino

GND → GND on Arduino

SCL → A5 (or SCL pin) on Arduino

SDA → A4 (or SDA pin) on Arduino

16x2 LCD with I2C Module to
Arduino Uno:

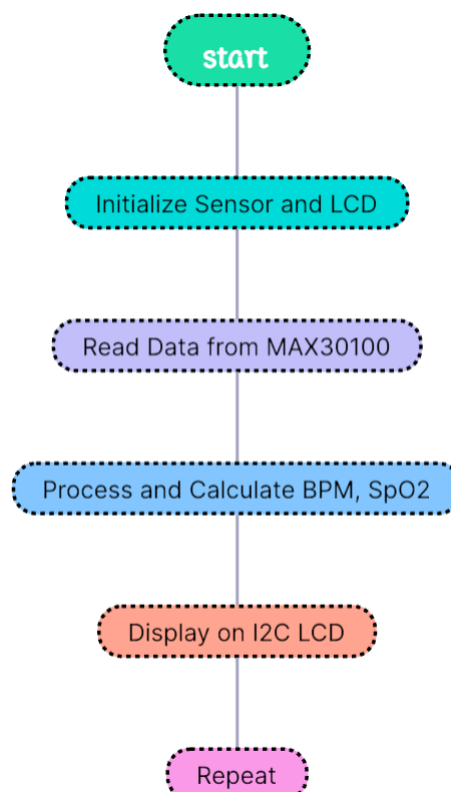
VCC → 5V on Arduino

GND → GND on Arduino

SDA → A4 (or SDA pin) on Arduino

SCL → A5 (or SCL pin) on Arduino

4.3 Flowchart (Basic Flow)





Chapter 5

Implementation

5.1 Code Outline

```
#include <Wire.h>
#include "MAX30105.h"
#include <LiquidCrystal_I2C.h>

MAX30105 sensor;
LiquidCrystal_I2C lcd(0x27,16,2);

float bpm = 0;
float spo2 = 0;

void setup() {
  lcd.init();
  lcd.backlight();
  sensor.begin(Wire, I2C_SPEED_STANDARD);
  sensor.setup(0x1F,4,2,100,411,4096);
}

void loop() {
  bpm = sensor.getHeartRate();
  spo2 = sensor.getSpO2();

  lcd.clear();
  lcd.setCursor(0,0);
  lcd.print("BPM:");
  lcd.print((int)bpm);
  lcd.print("%");
  lcd.setCursor(0,1);
  lcd.print("SpO2:");
  lcd.print((int)spo2);
  lcd.print("%");
  delay(1000);
}
```

5.2 Libraries Used

- #include <Wire.h>
- #include "MAX30105.h"
- #include <LiquidCrystal_I2C.h>

Chapter - 6

Results and Discussions

The IoT-based patient health monitoring system successfully measured and transmitted key health parameters such as heart rate, body temperature, and SpO₂ in real time. Sensor readings were accurate, with only minor deviations from standard medical devices. Data was sent to the cloud via Wi-Fi and displayed on a dashboard for continuous monitoring.

Alerts were triggered instantly when readings exceeded preset limits, and caregivers received notifications within seconds. The system showed low latency, good reliability, and efficient power consumption, making it suitable for remote monitoring.

Overall, the project proved that IoT technology can provide cost-effective, portable, and real-time patient health tracking. However, its performance depends on stable internet connectivity, and future improvements could include better sensor calibration, offline storage, and AI-based anomaly detection.

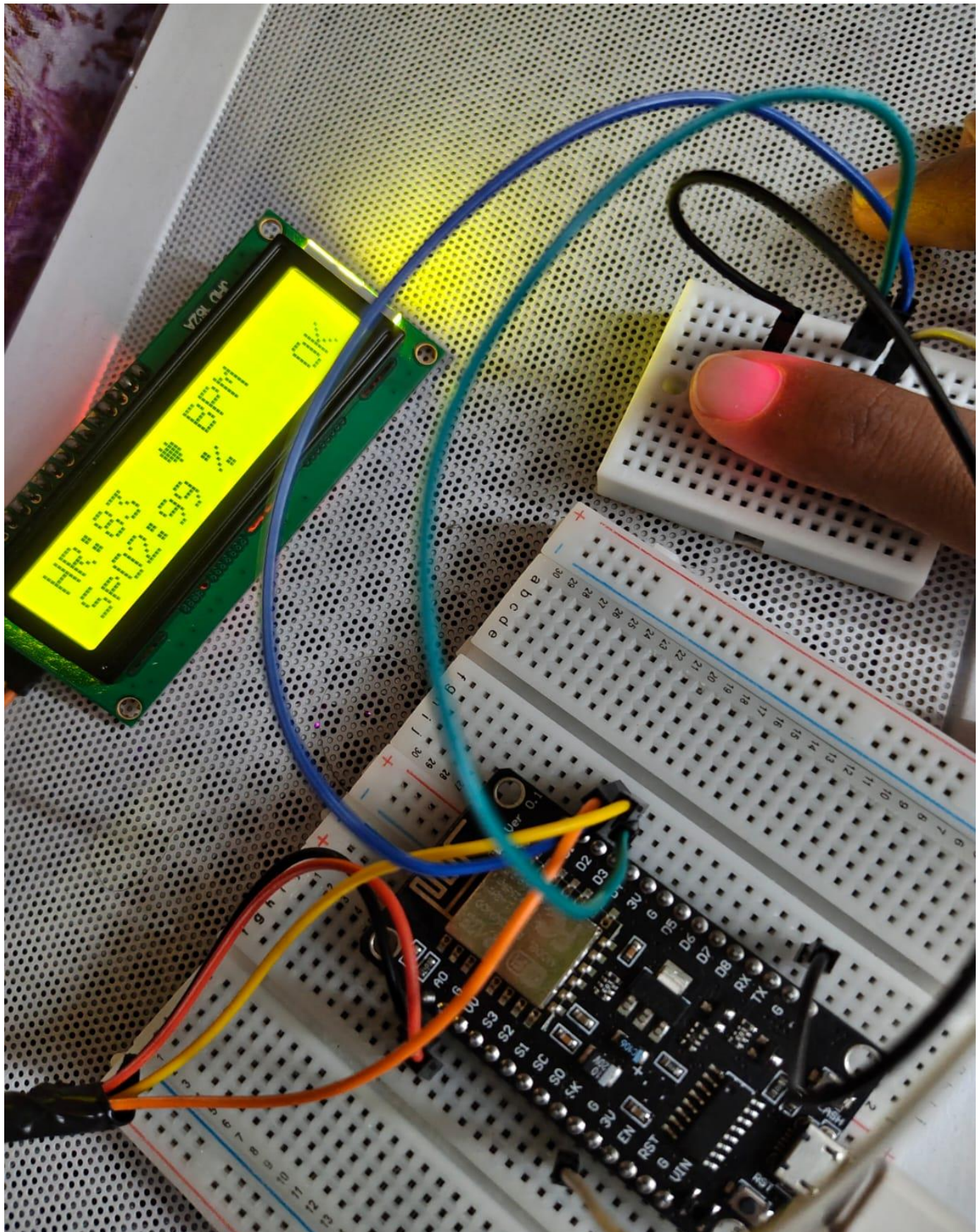


user interface



output screen

Project Image:





Chapter - 7

Conclusion & Future Scope

Conclusion

The IoT-Based Patient Health Monitoring System successfully monitors vital signs such as heart rate, and SpO₂ in real time. The system provides accurate readings, transmits data to the cloud, and sends alerts during abnormal conditions. It offers a low-cost, reliable, and portable solution for remote health tracking, especially useful for elderly or chronically ill patients. This project demonstrates how IoT can improve patient care, accessibility, and timely medical response.

Future Scope

The system can be enhanced by:

- Integrating additional sensors (ECG, blood pressure, glucose levels).
- Using AI and machine learning for health prediction and anomaly detection.
- Implementing data encryption to ensure patient data privacy and security.
- Enabling mobile app integration for better user interaction.
- Incorporating edge computing to reduce cloud dependency and improve response time.
- Adopting wearable technology for continuous, non-invasive monitoring.



REFERENCES

- [1] **MAX30100 Pulse Oximeter Datasheet – Maxim Integrated**
- [2] Arduino Documentation – <https://www.arduino.cc>
- [3] MAX3010x Arduino Library – GitHub - https://github.com/sparkfun/SparkFun_MAX3010x_Sensor_Library
- [4] Liquid Crystal I2C Library – https://github.com/johnrickman/LiquidCrystal_I2C
- [5] YouTube Reference - <https://youtu.be/1LqBvkHTJXU?si=uALP6emBGEiecPNA>
- [6] **Arefin, T.** and **Azad, A.** (2024), “Design and Implementation of an IoT Based Remote Health Monitoring System,” *Journal of Computer and Communications*, Vol. 12, No. 11, pp. 37-52.
- [7] **Lodi, E., Malto, M.M., and Verma, R.** (2023), “Measurement of BPM and Spo2 Using Max30102,” *International Journal of Innovative Science and Research Technology*, Vol. 8, Issue 5, pp. 1092-1096.
- [8] **Vasam, V.K., Shaik, A., Tolchuri, S.L.M., Rachamadugu, S.S., Krishna, H., Chandra, G.R., and Anand, D.** (2024), “Patient Health Monitoring System Using IOT And AI,” *Engineering Proceedings*, Vol. 66, No. 1, p. 31.
- [9] **Lin, M., Huang, Y., and Shih, Y.** (2024), “A Novel Algorithm for Detecting Pulse and Oxygen Saturation from MAX30102 PPG Signal,” *Journal of Biomedical Science and Engineering*, Vol. 17, No. 3, pp. 69-80.